VAULT · THE RIG · OCTOBER 2026

The Vault rig

A computer that works only while its owner's own voice and hands are producing what it receives, and every agent action traces back to them. Six pieces, five checks, all required, nothing on the wrists.

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Also the command chain · full research report

IN ONE PARAGRAPH

Six pieces, five checks, nothing on the wrists.

Every few seconds, Vault proves that the owner's own body is producing what the computer receives, and that every agent action traces back to it. **Earbuds** carry the voice and the ear. **A signed keyboard, a hands tracker and a faint signal sent through the body** carry the hands. **An IR camera** carries the face. **The hub** ties them together and holds the keys. **A master agent and a checking gateway** carry every command down to the tools. All five checks must hold. Two parts have to be built, the sensing earbuds and the body-sensing keycaps; the rest exists as products or developer kits.

Numbers are lab values from the cited studies, and every threshold has to be calibrated per owner. Nothing here has been tested end to end. "Proposal" marks design choices made for Vault rather than measured results.

01 · THE RIG

Six pieces, one job each

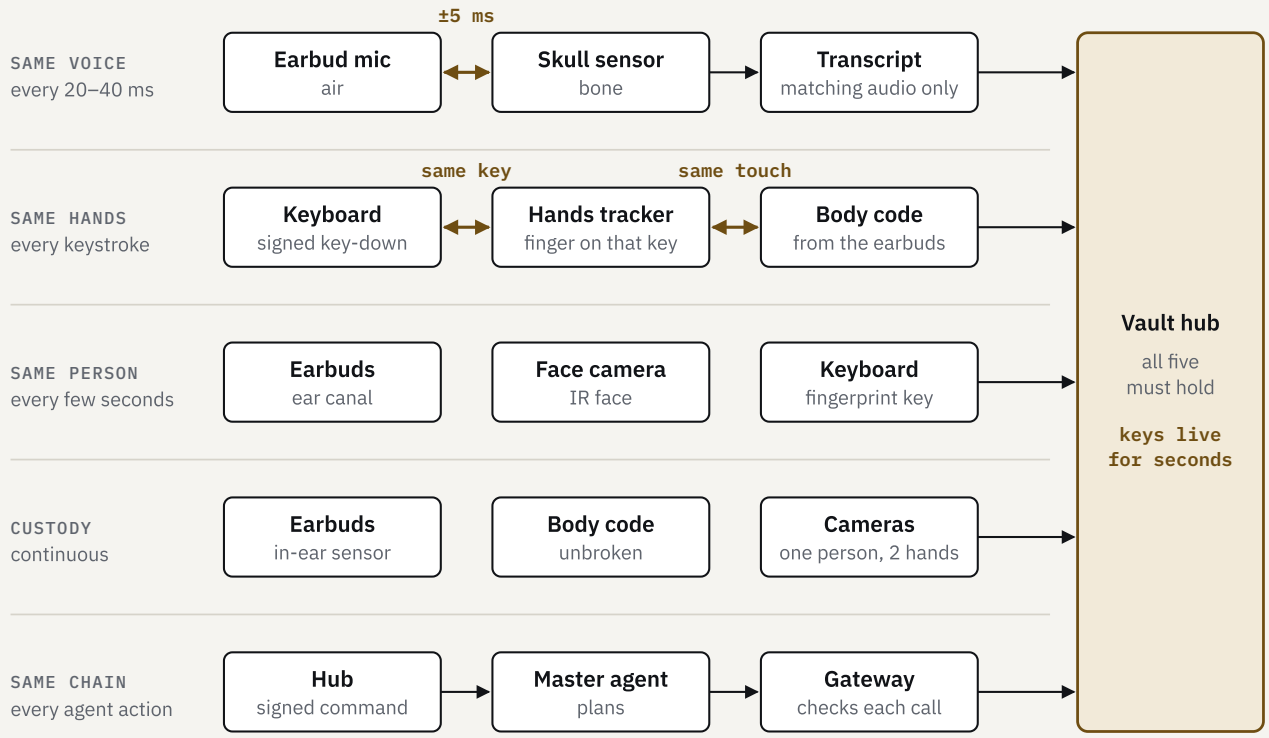
PIECE	WHERE	WHAT IT DOES	BUILD IT TODAY WITH
Wired earbuds	Both ears	Ear-canal echo says *who*; a skull-vibration sensor and inner mic prove *your* voice for dictation; an in-ear sensor detects removal; a faint coded signal goes out through your skin to the keys	Custom. OpenEarable 2.0 has the bone sensor and inner mics[1]; ear-canal ID reached 98.7% in research[2]; body-signal chips exist (Ixana Wi-R, vendor-claimed 5 Mb/s)[3]
Signed keyboard	Desk	Signs every key event with its own timestamp; a fingerprint key for identity, confirmations and a duress finger; keycaps or palm rest that receive the body signal	Wooting 80HE analog keyboard, $199.99[4], plus a fingerprint key paired the way Apple's Touch ID keyboard pairs[5]; the body-signal receiver is custom
Hands tracker	Above the keyboard, looking down	Infrared camera at 120 fps that sees which finger lands on which key	Ultraleap Leap Motion Controller 2: 120 fps, 850 nm infrared[6], with a supported screen-top mount[7]
Face camera	On the monitor	Infrared face on a signed path that software cannot feed	Integrated IR camera, as in Windows Enhanced Sign-in Security[8], with signed frames
Vault hub	Desk, cabled to the computer	Runs the checks, keeps one clock, holds the keys, shows and reads confirmations, keeps the trace log	Custom: a secure element with a small screen and speaker
Master agent and gateway	Software, keys held in the hub	Turns verified commands into signed roots and checks every agent action	Open building blocks: narrowing tokens[9], injection-resistant planning[10]; see the command chain

Everything is cabled to the hub, because Bluetooth relays survive encryption and add only about 8 ms[11]. The earbuds' cable also closes the body-signal loop through a shared ground. That is the DiamondTouch table's design in reverse: in 2001 it sent a signal through each user to a receiver in their chair, so it knew who touched where[12].

02 · FIVE CHECKS

Five checks, all required

Five checks run at once, and every piece feeds at least two of them, so faking or stealing one piece breaks another check.



How the pieces check each other. No piece is trusted on its own. Speech counts only when the skull sensor felt it[15]; a keystroke counts only when a fingertip was seen on that key and the body code came through it[22, 23]; an agent action counts only when it traces to a signed command[9, 10]. The hub keeps the computer's keys alive for a few seconds at a time, only while all five checks hold.

CHECK	QUESTION	HOW	STOPS	DOES NOT STOP
Same voice	Did the wearer's own throat produce this speech?	Each voiced 20–40 ms piece the earbud mic hears must also show up as skull vibration in the same earbud	Loudspeakers, AI voice clones, injected audio	The owner speaking under coercion

CHECK	QUESTION	HOW	STOPS	DOES NOT STOP
Same hands	Did hands attached to the earbud wearer press these keys?	Each signed key-down must match a fingertip landing on that key and arrive carrying the earbuds' body code	AI agents, malware, injected USB devices, another person typing	Someone touching the owner while typing, since the code passes through touch
Same person	Is this the owner?	Ear canal in speech pauses, IR face, fingerprint key	Another person wearing the earbuds	A coerced owner
Custody	Has anything changed since the owner last proved who they are?	Earbuds in, body code unbroken, one person in view, one pair of hands	Gear handed over mid-session; a second person joining	—
Same chain	Does every agent action trace to a verified command?	Hub-signed roots, tokens that only narrow, a gateway that checks every call	Prompt-injected or rogue agents acting on their own	A verified command that was itself a mistake

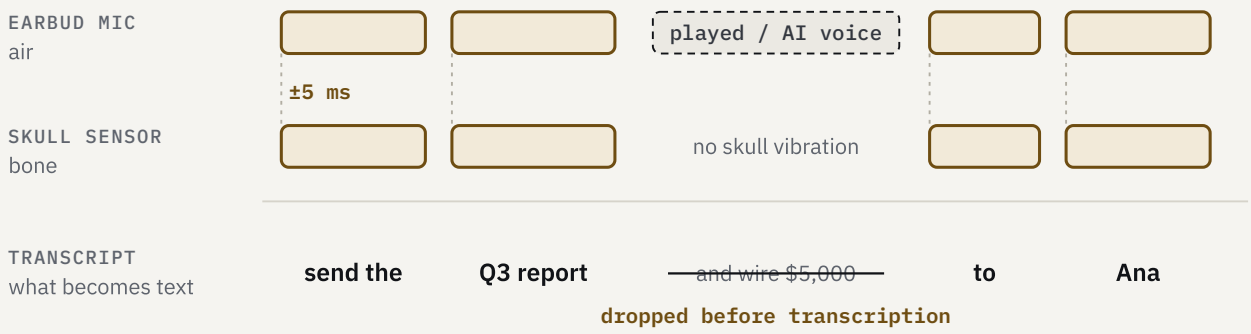
Each attack fails a different check. A voice clone fails *same voice*. An agent or a malicious USB cable fails *same hands*. Stolen earbuds fail *same person*. A swap mid-session fails *custody*. An agent tricked by a web page fails *same chain*. Only a coerced owner passes all five, which is why the fingerprint key also takes a duress finger.

03 · DICTATION

The same-voice check

AirPods already ship a "speech-detecting accelerometer"[13]. Apple's patent describes it as an in-canal sensor sampled at 2–6 kHz, used for noise suppression and closed to developers[14]. Research turns the same signal into a security check. VAuth accepted 97% of genuine commands with 0.09% false positives across 18 users. Airborne sound could fool it only from within about 30 cm at 90 dB[15]. AirBone reached a 1.1% equal error rate for liveness and 4.0–4.5% against replay and impersonation, with 31 users[16].

The check runs on the earbud itself (proposal). The skull sensor and inner mic share the earbud's clock and are signed by it. The physical lag between the two paths is under 1 ms, since the skull carries sound across the head in about 0.4 ms[17]. Every 20–40 ms of voiced speech must line up within ±5 ms, and only matching audio reaches transcription. Words get ±50 ms of slack, because even the best aligner places only about 73% of word boundaries within 25 ms[18]. Dictated commands become signed roots only from this verified audio.



Speech counts only if your skull felt it. The earbud's microphone hears everything in the room. The skull sensor in the same earbud feels only the wearer's own voice. Each voiced piece must appear in both, on the earbud's own clock, within a few milliseconds, or it never reaches transcription[15, 17]. The skull signal shows *when* you spoke precisely, and *what* you said only roughly[19].

Three limits remain:

- **What vs when.** The skull signal proves *when* you spoke with high confidence, but *what* you said only roughly. The best ear-worn systems recognize about 95% of words from a 100-word vocabulary[19]. Unvoiced sounds such as *s*, *f* and *t* barely reach the skull sensor[14].
- **Sensor forging.** Loud sound at a sensor's resonant frequency biased 75% of 20 tested accelerometer models[20]. The earbud must sign its data and filter out-of-band vibration.
- **Untested attacks.** No one has tested modern AI voice clones against this check, or speech injected while the owner is talking[15]. Jaw movement also changes the ear canal's shape, so ear-canal ID runs in speech pauses[19].

The same-hands check

Two signals replace the wristbands. The **hands tracker** watches your fingers from above. Vault doesn't need it to read what you type, because the keyboard already reports the key. It only asks yes or no: did a fingertip land on that key at that moment? Cameras already do the harder job well. Headset cameras detected touches at an F1 score of 0.99 and named the finger at 0.96, within 6.3 mm[21]. Phone video recovered typed text at a 0.3–1.1% character error rate from 0.8–3 m[22]. The weak spot is off-the-shelf fingertip tracking. Keys sit less than 1 cm apart, and one general-purpose tracker missed 7–10% of keystrokes[22]. So Vault judges windows of keys, not single keys (proposal).

The **body code** ties those hands to the earbuds. The earbuds send a faint coded signal at roughly 100 kHz–1 MHz, which spreads over the skin and is received by the keycaps or palm rest on every touch[23]. The current is a few microamps, about 2,000 times below the contact current where pain begins[24, 25]. Earlier security prototypes ran at 0.5–1 kb/s and needed touches over 300 ms to deliver a full 128-bit code[26]. Vault needs at least 10–100 kb/s, which body-network chips advertise[3]. The signal mostly stays on the body: an Oxford prototype told the body channel from fakes 99.3% of the time[27].

The honest limit is that the code proves the key was pressed by a body *in contact with* the earbud wearer, not by the wearer specifically. Anyone holding the owner's hand carries it too, and a live relay through a second person cannot be stopped by timing[28]. That's why the hands tracker also requires exactly one pair of hands, and the face camera exactly one person (proposal). Mimicking the owner's typing near a wrist-sensor check kept 40–45% of attackers logged in[29], which is why the hub, not the attacker's input, decides when comparisons run.

Commands and agents

Everything the verified owner says or types becomes a hub-signed command. A master agent plans only from those commands. It reads web pages, email and files through a quarantined reader that has no tools, and it hands sub-agents tokens that can only narrow[9, 10]. A gateway outside any AI model checks every action against its chain. It asks the owner when an action is consequential or was shaped by untrusted data, and logs everything, including refusals. The full design is on the command chain page.

06 · IF A PIECE IS MISSING

If a piece is missing

Missing data lets trust fade for a few seconds. Contradicting data locks at once: a different face, a key with no fingertip or body code, speech with no skull vibration (proposal).

PIECE OFF OR FAILING	WHAT GOES BLIND	WHAT VAULT DOES (PROPOSAL)
Earbuds out	Voice, ear identity, body code	Typing and dictation stop reaching the computer at once. When the earbuds go back in, the fingerprint key and face re-anchor
Hands tracker blocked	Same hands	Short gaps are tolerated. After a few seconds, keys stop reaching vault actions until the view is back
Face out of view	Face identity	Nothing for short gaps, while the ear check carries identity. Vault actions wait for the face
A second person in view or touching the keyboard	Custody	Vault actions pause until the owner confirms with the fingerprint key
Unknown keyboard or USB device	Signed input	Its input is blocked
Hub unplugged	Everything	Keys expire within seconds; the computer locks

07 · OPTIONS SCORED

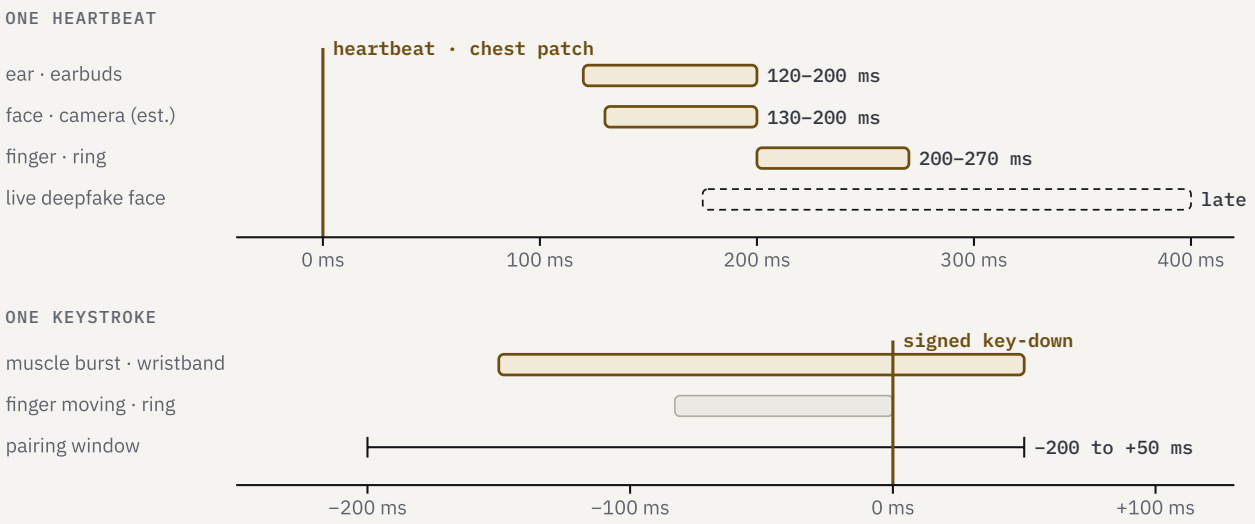
The options, scored

OPTION	VERDICT	WHY
Wired earbuds	**Must-have**	Voice binding for dictation, all-day ear identity, and the carrier of the body code. The cable rules out radio relay
Signed keyboard with fingerprint key	**Must-have**	Every key event signed at the switch. Identity on demand, the confirm button and a duress finger
Hands tracker (new)	**Must-have**	Replaces the wristbands. Checks that a finger landed on each reported key
Body-signal keycaps (new)	**Must-have, to be proven**	Ties each key to the body wearing the earbuds
IR face camera on an attested path	**Must-have**	The owner's face whenever they face the screen, over a path software cannot feed[8]
Master agent and gateway	**Must-have**	Every agent action traced to a verified command
sEMG wristbands	**In reserve**	The strongest per-key evidence, since muscles fire about 150 ms before a key bottoms out[30]. Use them only if the hands tracker and body code fail testing
Chest or microneedle patch	**Optional**	A single-use custody seal for the highest-stakes sessions[31]
Heartbeat timing check	**In reserve**	Needed only if Vault must accept ordinary webcams; see section 8
Ring, watch, Apple Watch ranging	**Drop**	Wrist pulse fails as identity in real use[32]. Watch ranging proves a token is near, not who is typing[33]
Nymi-style ECG band	**Drop**	Replayed ECG fooled it 81% of the time[34]
RGB webcam face	**Demote**	Never identity, because virtual cameras and injected frames can feed it[35, 36]
DNA or metabolome	**Drop**	Rapid DNA takes about 90 minutes[37]; metabolic profiles re-identify only about 40% of people after 7 years[38]

08 · IN RESERVE

Muscle and heartbeat timing

These two checks were the earlier answer to "is a human typing?" and "is that face real?". They stay in reserve.



One heartbeat and one keystroke, in milliseconds. Top: seated-adult values measured from the heartbeat. The face value is an estimate, because no study gives it directly. A check would compare the *gaps* between sites, calibrated per owner, over 10–30 beats[40, 41]. A live deepfake's face pulse comes from the attacker's own heart and lands late, because rendering adds at least ~43 ms per frame[35, 43]. Bottom: wrist-muscle activity starts about 150 ms before the key bottoms out, and the finger starts moving about 83 ms before. Every signed key-down must pair with one burst inside the window[30].

Muscle to keystroke. Forearm muscles fire about 150 ms before a key bottoms out, and a detector can place a keystroke to within about 21 ms[30]. Even a decoder trained on its owner misreads 7–11% of characters, so decisions run over 20–50 keys[39]. This is the strongest evidence that a specific hand pressed a key, but it requires wristbands.

Heartbeat timing. After each heartbeat, the pulse reaches the ear about 120–200 ms later and the finger about 200–270 ms later[40, 41]. Successive beats differ by about 42 ms, so a forger can't predict them[42]. A live deepfake's face pulse comes from the attacker and arrives late, because rendering adds at least ~43 ms per frame[35, 43]. With a camera that signs its frames, this check is redundant.

09 · NOT PROVEN YET

What is not proven yet

- **None of the three new checks has been measured.** That covers skull-versus-mic matching against modern voice clones, a hands tracker matched to signed key events, and ear-to-fingertip body signalling. Every number above comes from a separate lab study.
- **Two parts do not exist as products.** Wired earbuds with a skull sensor, inner mic and body-signal transmitter, and keycaps or a palm rest that receive the signal.
- **Touch passes the body code.** A second person touching the owner carries it, and a live relay cannot be timed out. The one-person and one-pair-of-hands rules carry that risk.
- **All-day earbuds.** Sealed earbuds bring pressure, heat and the muffled sound of your own voice. No study of consumer earbuds worn for 8+ hours was found.
- **Coercion.** All five checks pass for a coerced owner. The duress finger, delays and second approvers handle that; see the full report.
- **The agent chain's open problems.** No planner design has been tested against adaptive attackers, and turning speech into permissions is itself a trust boundary; see the command chain.

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